Manuscript Outline & Key Points

Nucleotide-resolution prediction of ribosome occupancy from transcriptome sequences across cellular contexts

INTRODUCTION

- Protein output cannot be inferred reliably from transcript abundance alone because translation varies across transcripts and cellular contexts. Ribosome profiling has revealed translated regions in canonical coding transcripts and in many RNAs previously classified as noncoding. These observations expand the biological scope of translation and motivate predictive models that resolve where ribosomes occupy each transcript. Such models could support both sequence design and the discovery of translated open reading frames (ORFs).
- Ribosome profiling measures ribosome occupancy at nucleotide resolution, but its experimental requirements limit coverage across tissues, cell states and clinical samples. Existing computational methods address parts of this problem. Some estimate transcript-level translation efficiency or local CDS density, whereas others identify ORFs from Ribo-seq or predict translation boundaries from sequence. However, these capabilities are usually separated across models, and several approaches require CDS annotations, fixed sequence windows or experimental Ribo-seq inputs. A unified model must predict a full-length occupancy profile while remaining annotation-free at inference.
- This task is challenging for three reasons. First, translation depends on sequence features ranging from local initiation motifs to full-transcript architecture. Second, the same RNA can be translated differently across cellular contexts. Third, training data from different Ribo-seq studies vary in coverage, protocol and P-site assignment. Addressing these challenges requires a harmonized dataset, a variable-length sequence model and explicit cellular context conditioning. It also requires evaluation on held-out transcripts that do not overlap the training set.
- Here we present TRACE (Translation Resolution Across Cell Environments), a transformer that predicts a nucleotide-resolution ribosome occupancy profile from full-length RNA sequence and cellular context. TRACE was trained on 1.8 million transcript–context profiles from 73 cell types across three species. Among the methods compared in Supplementary Table 1, only TRACE combines full-length variable-length input, cellular context conditioning, unified multi-species modeling, nucleotide-resolution occupancy prediction and annotation-free inference. This translation representation supports tasks that are commonly separated. Within a known CDS, occupancy magnitude and position provide an objective for forward sequence design. Across an unannotated transcript, the same profile supports reverse discovery by localizing candidate translated ORFs. We evaluate these capabilities across translation dynamics, translation efficiency, ORF identification, cultured-cell mRNA design and tumor neoantigen candidate prioritization.

RESULT 1: A nucleotide-resolution dataset of ribosome occupancy across cellular contexts

- Ribo-seq signal is confounded by transcript abundance and experimental protocol variability; raw counts do not reflect intrinsic translational capacity.
- Curated 1,355 Ribo-seq runs and 475 matched RNA-seq runs from 73 cell types across human, rhesus macaque and mouse.
- Built a unified pipeline: quality filtering → merging same-cell-type data → transcript-level depth/coverage filtering → LightGBM-based P-site prediction → periodicity filtering (>0.5).
- Merging same-cell-type data substantially improved per-transcript coverage; CDS coverage of housekeeping genes approached 1.0 after merging.
- Applied compositional regression to reduce RNA-abundance confounding and generate nucleotide-resolution ribosome occupancy profiles.
- Between-species and cross-context patterns are consistent with a substantial contribution of cis sequence, but do not establish causal dominance. Related cellular contexts still cluster by translation efficiency across species. This motivates an RNA-sequence backbone with cellular context conditioning.
- The unified pipeline transformed heterogeneous raw signals from disparate studies into coherent profiles: translation signal became sharply enriched around CDS start sites, with strong three-nucleotide periodicity across the CDS body.
- Final dataset: 1.8 million transcript–context profiles, with one profile per transcript and available cellular context (Fig. 1).
- The dataset includes canonical protein-coding transcripts and noncoding RNAs with measurable ribosome occupancy; it is not claimed to represent every translatome state.

RESULT 2: TRACE predicts nucleotide-resolution ribosome occupancy profiles

- The TRACE backbone uses full-length RNA sequence as a shared representation, consistent with a substantial cis-sequence contribution. Bidirectional attention, Rotary Position Embedding, Flash Attention and length-aware batching support variable-length transcripts.
- Adaptive Layer Normalization introduces cellular context conditioning throughout the sequence backbone rather than treating context as a separate task-specific model.
- Training uses position-level density, a CDS frame-aware TE objective and a transcript-ranking objective. CDS boundaries and reading frames provide training supervision; codon identity is represented through the input sequence, and RBP motif labels are not supplied. TRACE remains annotation-free at inference.
- On chromosome-held-out transcripts, annotation-free inference produces periodic, CDS-enriched occupancy profiles; this reflects generalization of supervised translation patterns (Fig. 2b,c).
- Predicted periodicity differs between housekeeping transcripts and long noncoding RNAs, with the supervision boundary stated explicitly (Fig. 2d).

- Example: ENST00000332859 reaches Spearman ρ = 0.80; a localized prediction on noncoding RNA ENST000000789734 overlaps experimental Ribo-seq evidence without establishing stable protein production.
- One nucleotide-resolution ribosome occupancy profile serves as a unified translation representation. Known-CDS aggregation supports TE estimation and forward sequence design; annotation-free scanning supports reverse ORF discovery.

RESULT 3: A unified translation representation supports three benchmark tasks

- TRACE is evaluated on translation dynamics, TE and ORF identification. Within the models summarized in Supplementary Table 1, only TRACE combines nucleotide-resolution occupancy, explicit cellular context, unified multi-species modeling and annotation-free inference.
- Translational dynamics: TRACE approaches cross-study experimental baselines and exceeds retrained sequence-only Riboformer and RiboMIMO in the evaluated datasets (Fig. 3b–d).
- Translation efficiency: TRACE ranks among the top-performing models by Spearman ρ for polysome profiling, protein-to-RNA ratios and SILAC synthesis rates (Fig. 3e–g).
- ORF identification: TRACE attains higher F1 than the evaluated Ribo-seq-based callers and sequence baselines without sample-specific Ribo-seq input.
- Matched fetal-brain Iso-Seq, Ribo-seq and mass spectrometry: TRACE profiles correlate with Ribo-seq; short-ORF precision is comparable to RiboTIE.

RESULT 4: TRACE predictions show translation-associated patterns across sequence scales

- Prediction sensitivity and attention are analyzed across transcript architecture, initiation motifs and codon positions. Because CDS and frame labels contributed to training, attention is treated as an associative explanation rather than causal evidence.
- Predicted TE is associated with 5′ UTR, CDS and 3′ UTR length and with RNA minimum free energy; these correlations do not establish causal effects.
- Attention is enriched near start, CDS and stop regions and is associated with Kozak context; these patterns are reported as model associations.
- Frame-resolved attention reflects training supervision. In silico codon substitutions probe prediction sensitivity rather than establish an elongation mechanism.
- In silico upstream-start perturbations produce prediction changes consistent with known initiation-associated patterns; the interpretation remains model based.

RESULT 5: Forward design of CDS translation in specific cellular contexts

- Ten Fluc and mCherry designs show graded cultured-cell protein output associated with TRACE predictions (Pearson r > 0.9).

- A TRACE-designed Gluc CDS produces higher measured expression than wild-type, RiboDecode and LinearDesign sequences in the evaluated cultured-cell experiment.
- Cellular-context-specific reporter designs produce the intended relative expression patterns in HEK293T, HeLa and HepG2 cells.

RESULT 6: Reverse discovery of tumor neoantigen candidates from transcriptomes

- Non-canonical translation and alternative splicing provide candidate epitopes beyond somatic mutations, but require prioritization of potentially translated ORFs.
- Annotation-free occupancy prediction supports candidate nomination from tumor/normal RNA-seq without matched tumor Ribo-seq or mass spectrometry.
- Patient pipeline: tumor/normal RNA-seq → isoform assembly → tumor-specific transcripts → TRACE ORF prediction → HLA binding prediction → ten candidate neoepitopes.
- IFN-γ ELISpot and organoid co-culture provide functional support for selected candidates but do not directly quantify endogenous peptide abundance.

DISCUSSION

- A harmonized dataset of 1.8 million transcript–context profiles supports nucleotide-resolution prediction across 73 cell types and three species.
- TRACE uses full-length RNA sequence as the backbone and cellular context as conditioning.
- Unified application logic: position-level CDS occupancy supports forward design, whereas the same annotation-free profile supports reverse discovery of candidate ORFs and neoantigens.
- Limitations: occupancy is not equivalent to stable protein; low-density and underrepresented contexts remain difficult; functional neoantigen assays do not replace direct immunopeptidomic confirmation.
- Near-term extensions include incorporating RNA structure, RNA modifications, RBP occupancy and subcellular localization, improving generalization to sparsely sampled cellular contexts, and obtaining direct immunopeptidomic validation of candidate neoantigens.
- More broadly, the harmonized data representation and joint training strategy developed here may provide a translation-focused adaptation framework for RNA foundation models. Fine-tuning such models with nucleotide-resolution ribosome occupancy profiles, ORF supervision and transcript-level TE objectives could enable a shared pretrained RNA representation to support both ORF identification and TE quantification.
- TRACE may also provide a sequence-design objective for cell-type-specific protein expression. Potential applications include optimizing mRNAs for protein-replacement therapies and tuning the expression of Cas and other genome-editing proteins. These applications remain to be tested in vivo, where delivery, RNA stability, innate immune activation and tissue-specific regulation may alter the relationship between predicted ribosome occupancy and therapeutic protein output.

- Finally, applying annotation-free TRACE predictions to population-scale tumor transcriptomes could help identify recurrent, tumor-associated translated ORFs and candidate neoepitopes shared across patients or population groups. If validated by immunopeptidomics and functional immune assays across diverse HLA backgrounds, such candidates could support more broadly deployable mRNA vaccines and reduce, rather than eliminate, the cost and turnaround time of fully individualized antigen discovery and vaccine design.

FIGURES

Figure	Content
Fig. 1	Dataset: nucleotide-resolution ribosome occupancy and cellular context-associated patterns
Fig. 2	TRACE architecture, annotation-free inference on held-out transcripts, CDS enrichment and biotype periodicity
Fig. 3	Unified benchmarks: translation dynamics, TE, ORF identification and matched multi-omics validation
Fig. 4	Associative interpretation: architecture, MFE, attention, Kozak context, codon-position sensitivity and upstream starts
Fig. 5	Forward design: graded reporters, Gluc comparison and expression specific to each cellular context in cultured cells
Fig. 6	Reverse discovery: neoantigen candidate prioritization, IFN-γ ELISpot and organoid co-culture